Machine Learning in Finance and Insurance

Coding Projects 2026

ETH zürich

D MATH

General Info

- **Questions** can be posed on the Moodle Q&A Forum

- **Projects Schedule**

Project	First discussion	Last questions	Submission deadline
Linear Regression	Sept 23	Oct 7	Oct 14
Credit Analytics	Oct 14	Oct 28	Nov 4
Deep Hedging	Nov 4	Nov 18	Nov 25
Insurance Claims Prediction	Nov 25	Dec 9	Dec 16

Project Submission

- Projects are guided coding exercises in which a topic from the lectures is implemented
- Project material will be available on Moodle
- You can work in teams of up to 5 students:
  - Teammates can be found via the Moodle Forum “Find Your Teammates"
  - You can join a different team for each coding project
- Projects must be coded in Python
- Submit a **fully executed Jupyter notebook**, including a concise report in **Markdown Cells** at the end of the notebook, under the heading **Project Report**.
- Write the names of all team members at the beginning of the notebook
- Submit only one notebook per team on Moodle
- You can adapt/use code from our sample Jupyter notebooks
- Projects are graded and account for 30% of the total grade

Project Grading

Code: 80%

- Correctness
- Implementation
- Quality of the results
- Model and hyperparameter selection
- Validation

Report: 20%

- Conciseness
- Readability
- Completeness

Project Report

- **Approach:** model or method used and main modelling choices
- **Model selection:** architecture and hyperparameter choices and how they were validated
- **Key results:** main quantitative results, comparisons and relevant figures or tables
- **Interpretation:** what the results mean and why the chosen model performs well
- **Robustness:** is the model robust with respect to the choice of the architecture and the parameters?
- **Limitations:** What are the limitations of the approach? Could one obtain better results with a different method?

Python Packages

- Python 3
  - Tutorial: https://docs.python.org/3/tutorial/index.html
  - Documentation: https://docs.python.org/3/library/index.html
- numpy (linear algebra and more...)
  - Tutorial: https://numpy.org/learn/
  - Documentation: https://numpy.org/doc/stable/
- matplotlib (data visualization)
  - Tutorial: https://matplotlib.org/stable/tutorials/index
  - Documentation: https://matplotlib.org/stable/api/index.html
- pandas (data manipulation)
  - Tutorial: https://pandas.pydata.org/docs/getting_started/index.html
  - Documentation: https://pandas.pydata.org/docs/reference/index.html
- SciPy (numerics)
  - Documentation: https://docs.scipy.org/doc/scipy/reference/index.html

Python ML Packages

- sklearn (regression models, SVMs, and many more ...)
  - Tutorial: https://scikit-learn.org/stable/getting_started.html
  - Documentation: https://scikit-learn.org/stable/modules/classes.html
- statsmodels (estimation and inference of statistical models)
  - Tutorial: https://www.statsmodels.org/stable/gettingstarted.html
  - Documentation: https://www.statsmodels.org/stable/index.html
- PyTorch (neural networks)
  - Tutorial: https://pytorch.org/tutorials/beginner/basics/intro.html
  - Documentation: https://pytorch.org/docs/stable/index.html
- TensorFlow (neural networks)
  - Tutorial: https://www.tensorflow.org/tutorials
  - Documentation: https://www.tensorflow.org/api_docs/python/tf
- Keras (a TensorFlow API)
  - Tutorial: https://keras.io/getting_started/
  - Documentation: https://keras.io/api/

Jupyter Notebooks

- Install Jupyter locally on your laptop
  - Tutorial: https://jupyter.org/install
  - Documentation: https://docs.jupyter.org/en/latest/
- Use Jupyter online:
  - Google Colab: https://colab.google/
  - JupyterLab: https://jupyter.org/try-jupyter/lab/